

Explainable Artificial Intelligence (XAI)

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Transparency is described in the article which states that it is “the search for a direct understanding of the mechanism by which a model works” (p. 83). Essentially, it is the search for a deeper and clearer understanding of black-box models that are inherently hard to decipher, especially if the user is from a non-technical background. Additionally, explainability is defined in the article as placing the audience at the forefront of understanding when considering a machine learning model. The two concepts therefore coincide with each other by stating that we need to create a better understanding of how machine learning models work and require us to have an audience in mind when creating an explanation. Thus, I would advocate for bringing these aspects into an ML/AI system to make sure that everyone involved with the model, whether that be developers or more importantly the users, are able to understand how their data is being used.

For users, introducing transparency and explainability can help with trustworthiness. Through explanations, user comprehension on why a system made a certain decision can increase and make them more willing to rely on the model. Additionally, by introducing more informativeness and clarity, users are much less likely to misuse the model, whether that is through proper confusion or actual ill-will. Following that, explanations can help reveal whether decisions were made on inappropriate or harmful factors which would allow users to detect bias and correct the issue.

For organizations, an important factor when considering transparency and explainability are regulatory compliance laws. Rules such as the GDPR require full transparency and the ability to explain why certain automated decisions were made. Additionally, elevating understanding of a model can aid in attempts to continue to develop systems in lieu of weaknesses, adversarial vulnerabilities (cyber security issues) or even pure data problems. Transparency/explainability can also help with holding organizations accountable while also giving them protection. If model decisions were previously explained and accepted, then when strange results come up there is an explanation provided which can give even more insight. Perhaps undiscovered latent findings could surface in these cases. Through transparency/explainability, organizations can also gain customer confidence and introduce even more adoption of the system.

From the article, a Fair AI is defined as a system that “should not lead to any kind of discrimination against individuals or collections in relation to race, religion, gender, [etc...]” (p. 103). In essence, it is a system whose decisions are not influenced by irrelevant,

inappropriate, or discriminatory factors. As such, a fair system should avoid bias and treat similar individuals similarly.

Take for example a resume AI system that vets job applications. Fairness in this model would be to make sure that only relevant factors are considered. These could include prior work history, education level, skills and coursework, etc. A fair system would not consider, nor be indirectly influenced by sensitive attributes such as race, gender, age, appearance, or location of residence that could encode demographic information. Explainability techniques could help show which features are contributing most heavily, searching for relevance. Importantly, identical applicants with similar relevant characteristics are evaluated identically.

Bias is the systematic, unfair errors or correlations that are produced from a model via a dataset. Fairness, in turn, is the goal of removing those unfair patterns so decisions made afterwards are not discriminative. Transparency (or explainability) helps to make clear where biases could come from, which features matter the most, and how decisions are made in the end; without transparency, fairness cannot be efficiently evaluated. Accountability is the responsibility of either the developer or organization to ensure that the model behaves correctly and ethically. With accountability, this aspect depends on both transparency and fairness. Transparency will help explain what a model is doing, and fairness will ensure it treats the clients or users equally. In short, transparency of a model is introduced which gives way for biases to surface. Fairness is then used to quell the biases which in turn supports accountability.

However, a system that is biased in the technical sense while still having fairness in the ethical sense can still exist. For example, a weather prediction model for a specific region could have technical biases while maintaining ethical fairness. The biases in this case would be past data from the specific region. Perhaps in Seattle during the fall season, temperatures consistently fall within a certain range. This information could heavily bias or influence future predictions. However, this statistical bias affects all residents equally. All residents in Seattle or nearby areas like Bellevue or Tacoma will most likely get similar weather conditions, regardless of factors like race, gender, or socioeconomical statuses. In this case the system is statistically biased but ethically fair.